Status	Finished
Started	Thursday, 8 October 2026, 12:56 PM
Completed	Thursday, 8 October 2026, 1:01 PM
Duration	4 mins 52 secs
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Write a Python program for binary search.

For example:

Input	Result
1,2,3,5,8 6	False
3,5,9,45,42 42	True

Answer: (penalty regime: 0 %)

```
import sys

def binary_search(arr, target):
    # Binary search requires a sorted array
    arr.sort()

    low = 0
    high = len(arr) - 1

    while low <= high:
        mid = (low + high) // 2
        if arr[mid] == target:
            return True
        elif arr[mid] < target:
            low = mid + 1
        else:
            high = mid - 1

    return False

def main():
    # Read all lines from standard input
    input_data = sys.stdin.read().splitlines()
    if not input_data:
        return

    # Process pairs of lines (Line 1: Array, Line 2: Target)
    for i in range(0, len(input_data), 2):
        if i + 1 >= len(input_data):
            break

        # Parse the comma-separated list of numbers
        arr = [int(x) for x in input_data[i].split(',') if x.strip()]
        # Parse the target integer
        target = int(input_data[i+1].strip())

        # Print ONLY the exact result expected (True/False)
        print(binary_search(arr, target))

if __name__ == '__main__':
    main()
```

	Input	Expected	Got	
✓	1,2,3,5,8 6	False	False	✓
✓	3,5,9,45,42 42	True	True	✓
✓	52,45,89,43,11 11	True	True	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **2**

Correct

Mark 1.00 out of 1.00

Balanced strings are those that have an equal quantity of 'L' and 'R' characters.

Given a balanced string s, split it in the maximum amount of balanced strings.

Return the maximum amount of split balanced strings.

Example 1:

Input:

RLRRLLRLRL

Output:

4

Explanation: s can be split into "RL", "RRLL", "RL", "RL", each substring contains same number of 'L' and 'R'.

Example 2:

Input:

RLLLLRRRLR

Output:

3

Explanation: s can be split into "RL", "LLLRRR", "LR", each substring contains same number of 'L' and 'R'.

Example 3:

Input:

LLLLRRRR

Output:

1

Explanation: s can be split into "LLLLRRRR".

Constraints:

1 <= s.length <= 1000

s[i] is either 'L' or 'R'.

s is a balanced string.

For example:

Test	Result
`print(BalancedStrings('RLRRLLRLRL'))`	4
`print(BalancedStrings('RLLLLRRRLR'))`	3

Answer: (penalty regime: 0 %)

Reset answer

```
def BalancedStrings(s: str) -> int:
    balanced_count = 0
    balance = 0

    for char in s:
        if char == 'R':
            balance += 1
        else:
            balance -= 1

        # A balance of 0 means the number of 'R's and 'L's match perfectly
        if balance == 0:
            balanced_count += 1

    return balanced_count

```

	Test	Expected	Got	
✓	`print(BalancedStrings('RLRRLLRLRL'))`	4	4	✓

	Test	Expected	Got	
✓	`print(BalancedStrings('RLLLLRRRLR'))`	3	3	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **3**

Correct

Mark 1.00 out of 1.00

Given an list, find peak element in it. A peak element is an element that is greater than its neighbors.

An element a[i] is a peak element if

A[i-1] <= A[i] >=a[i+1] for middle elements. [0<i<n-1]

A[i-1] <= A[i] for last element [i=n-1]

A[i]>=A[i+1] for first element [i=0]

Input Format

The first line contains a single integer n , the length of A .
The second line contains n space-separated integers,A[i].

Output Format

Print peak numbers separated by space.

Sample Input

5

8 9 10 2 6

Sample Output

```
10 6
```

For example:

Input	Result
4 12 3 6 8	12 8

Answer: (penalty regime: 0 %)

```
import sys

def find_peaks(n, arr):
    peaks = []

    # Handle single-element array edge case
    if n == 1:
        return [arr[0]]

    for i in range(n):
        # Case 1: First element
        if i == 0:
            if arr[i] >= arr[i + 1]:
                peaks.append(arr[i])
        # Case 2: Last element
        elif i == n - 1:
            if arr[i] >= arr[i - 1]:
                peaks.append(arr[i])
        # Case 3: Middle elements
        else:
            if arr[i] >= arr[i - 1] and arr[i] >= arr[i + 1]:
                peaks.append(arr[i])

    return peaks

def main():
    # Read all input from standard input
    input_data = sys.stdin.read().split()
    if not input_data:
        return

    # Parse n and the array elements
    n = int(input_data[0])
    arr = [int(x) for x in input_data[1:n+1]]

    # Find the peak numbers
    peaks = find_peaks(n, arr)

    # Print the peaks separated by space
    print(*(peaks))

if __name__ == '__main__':
    main()

```

	Input	Expected	Got	
✓	7 15 7 10 8 9 4 6	15 10 9 6	15 10 9 6	✓
✓	4 12 3 6 8	12 8	12 8	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **4**

Correct

Mark 1.00 out of 1.00

Given an array `nums` containing `n` distinct numbers in the range `[0, n]`, return *the only number in the range that is missing from the array.*

Example 1:

```
Input: nums = [3,0,1]
Output: 2
Explanation: n = 3 since there are 3 numbers, so all numbers are in the range [0,3]. 2 is the missing number in the range since it does not appear in nums.
```

Example 2:

```
Input: nums = [0,1]
Output: 2
Explanation: n = 2 since there are 2 numbers, so all numbers are in the range [0,2]. 2 is the missing number in the range since it does not appear in nums.
```

Example 3:

```
Input: nums = [9,6,4,2,3,5,7,0,1]
Output: 8
Explanation: n = 9 since there are 9 numbers, so all numbers are in the range [0,9]. 8 is the missing number in the range since it does not appear in nums.
```

For example:

Test	Result
`print(missingNumber([3,0,1]))`	2
`print(missingNumber([0,1]))`	2

Answer: (penalty regime: 0 %)

Reset answer

```python
def missingNumber(nums: list[int]) -> int:
    n = len(nums)
    # Calculate the expected sum of numbers from 0 to n
    expected_sum = (n * (n + 1)) // 2
    # Calculate the actual sum of elements in the array
    actual_sum = sum(nums)

    # The difference is the missing number
    return expected_sum - actual_sum

```

	Test	Expected	Got	
✓	`print(missingNumber([3,0,1]))`	2	2	✓
✓	`print(missingNumber([0,1]))`	2	2	✓
✓	`print(missingNumber([9,6,4,2,3,5,7,0,1]))`	8	8	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **5**

Correct

Mark 1.00 out of 1.00

Given an array of integers `nums` which is sorted in ascending order, and an integer `target`, write a function to search `target` in `nums`. If `target` exists, then return its index. Otherwise, return `-1`.

You must write an algorithm with `O(log n)` runtime complexity.

Example 1:

```
Input: nums = [-1,0,3,5,9,12], target = 9
Output: 4
Explanation: 9 exists in nums and its index is 4
```

Example 2:

```
Input: nums = [-1,0,3,5,9,12], target = 2
Output: -1
Explanation: 2 does not exist in nums so return -1
```

Constraints:

- `1 <= nums.length <= 10^4`
- `-10^4 < nums[i], target < 10^4`
- All the integers in `nums` are **unique**.
- `nums` is sorted in ascending order.

For example:

Test	Result
`print(search([-1,0,3,5,9,12],9))`	4

Answer: (penalty regime: 0 %)

Reset answer

```
def search(nums: list[int], target: int) -> int:
    low = 0
    high = len(nums) - 1

    while low <= high:
        # Calculate the middle index
        mid = (low + high) // 2

        # Target found, return its index
        if nums[mid] == target:
            return mid
        # If target is larger, search the right half
        elif nums[mid] < target:
            low = mid + 1
        # If target is smaller, search the left half
        else:
            high = mid - 1

    # Target does not exist in the array
    return -1

```

	Test	Expected	Got	
✓	`print(search([-1,0,3,5,9,12],9))`	4	4	✓
✓	`print(search([-1,0,3,5,9,12],2))`	-1	-1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **6**

Correct

Mark 1.00 out of 1.00

String should contain only the words are not palindrome.

Sample Input 1

Malayalam is my mother tongue

Sample Output 1

is my mother tongue

Answer: (penalty regime: 0 %)

```
s = input().lower()
print(' '.join(w for w in s.split() if w != w[::-1]))
```

	Input	Expected	Got	
✓	Malayalam is my mother tongue	is my mother tongue	is my mother tongue	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **7**

Correct

Mark 1.00 out of 1.00

An list contains N numbers and you want to determine whether two of the numbers sum to a given number K. For example, if the input is 8, 4, 1, 6 and K is 10, the answer is yes (4 and 6). A number may be used twice.

Input Format

The first line contains a single integer n , the length of list

The second line contains n space-separated integers, list[i].

The third line contains integer k.

Output Format

Print Yes or No.

Sample Input

7

0 1 2 4 6 5 3

1

Sample Output

Yes

For example:

Input	Result
5 8 9 12 15 3 11	Yes
6 2 9 21 32 43 43 1 4	No

Answer: (penalty regime: 0 %)

```
import sys

def has_two_sum(arr, k):
    # Use a set for O(1) lookups
    seen = set(arr)

    for num in arr:
        complement = k - num
        # Since a number can be used twice, we just check if the complement exists in the set
        if complement in seen:
            return "Yes"

    return "No"

def main():
    # Read all inputs from standard input
    input_data = sys.stdin.read().split()
    if not input_data:
        return

    # Parse inputs sequentially based on the described format
    n = int(input_data[0])
    arr = [int(x) for x in input_data[1:n+1]]
    k = int(input_data[n+1])

    # Print the result
    print(has_two_sum(arr, k))

if __name__ == '__main__':
    main()

```

	Input	Expected	Got	
✓	5 8 9 12 15 3 11	Yes	Yes	✓

	Input	Expected	Got	
✓	6 2 9 21 32 43 43 1 4	No	No	✓
✓	6 13 42 31 4 8 9 17	Yes	Yes	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **8**

Correct

Mark 1.00 out of 1.00

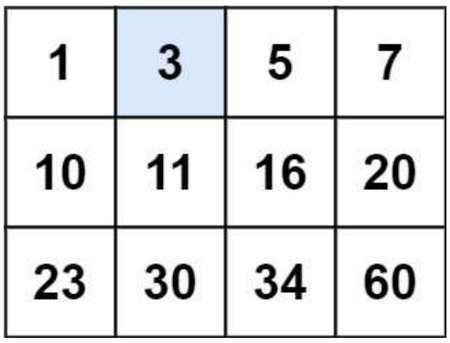
You are given an `m x n` integer matrix `matrix` with the following two properties:

- Each row is sorted in non-decreasing order.
- The first integer of each row is greater than the last integer of the previous row.

Given an integer `target`, return `True` *if* `target` *is in* `matrix` *or* `False` *otherwise.*

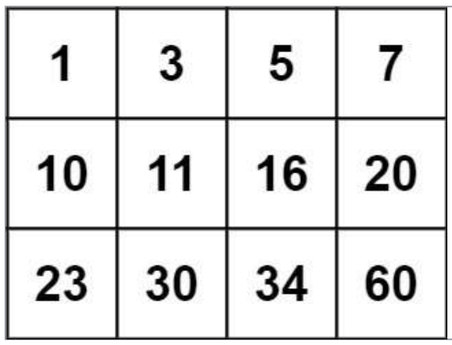
You must write a solution in `O(log(m * n))` time complexity.

Example 1:




```
Input: matrix = [[1,3,5,7],[10,11,16,20],[23,30,34,60]], target = 3
Output: True
```

Example 2:




```
Input: matrix = [[1,3,5,7],[10,11,16,20],[23,30,34,60]], target = 13
Output: False
```

For example:

Test	Result
`print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 13))`	`False`
`print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 3))`	`True`

Answer: (penalty regime: 0 %)

Reset answer

```python
def searchMatrix(matrix: list[list[int]], target: int) -> bool:
    if not matrix or not matrix[0]:
        return False

    m = len(matrix)       # Number of rows
    n = len(matrix[0])    # Number of columns

    low = 0
    high = (m * n) - 1

    while low <= high:
        mid = (low + high) // 2

        # Map the 1D index back to 2D coordinates
        row = mid // n
        col = mid % n

        mid_val = matrix[row][col]

        if mid_val == target:
            return True
        elif mid_val < target:
            low = mid + 1
        else:
            high = mid - 1

    return False
```

```
28
```

	Test	Expected	Got	
✓	print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 13))	False	False	✓
✓	print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 3))	True	True	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **9**

Correct

Mark 1.00 out of 1.00

Given two Strings s1 and s2, remove all the characters from s1 which is present in s2.

Constraints

1<= string length <= 200

Sample Input 1

experience
enc

Sample Output 1

xpri

Answer: (penalty regime: 0 %)

```
import sys

def remove_characters(s1: str, s2: str) -> str:
    # Convert s2 into a set for fast lookup
    chars_to_remove = set(s2)

    # Filter out characters present in s2 while keeping original order
    result = [char for char in s1 if char not in chars_to_remove]

    return "".join(result)

def main():
    # Read all lines from standard input
    input_data = sys.stdin.read().splitlines()
    if len(input_data) < 2:
        return

    s1 = input_data[0]
    s2 = input_data[1]

    print(remove_characters(s1, s2))

if __name__ == '__main__':
    main()

```

	Input	Expected	Got	
✓	experience enc	xpri	xpri	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **10**

Correct

Mark 1.00 out of 1.00

Two string values S1, S2 are passed as the input. The program must print first N characters present in S1 which are also present in S2.

Input Format:

The first line contains S1.
The second line contains S2.
The third line contains N.

Output Format:

The first line contains the N characters present in S1 which are also present in S2.

Boundary Conditions:

2 <= N <= 10
2 <= Length of S1, S2 <= 1000

Example Input/Output 1:

Input:

abcbde
cdefghbb
3

Output:

bcd

Note:

b occurs twice in common but must be printed only once.

Answer: (penalty regime: 0 %)

```
import sys

def print_common_characters():
    # Read all lines from standard input
    input_data = sys.stdin.read().splitlines()
    if len(input_data) < 3:
        return

    s1 = input_data[0]
    s2 = input_data[1]
    n = int(input_data[2].strip())

    # Store s2 characters in a set for O(1) lookups
    s2_chars = set(s2)

    # Track characters we have already added to avoid duplicates
    seen = set()
    result = []

    # Find the matching unique characters in the order they appear in s1
    for char in s1:
        if char in s2_chars and char not in seen:
            seen.add(char)
            result.append(char)
            # Stop once we've collected N unique common characters
            if len(result) == n:
                break

    # Print the resulting characters joined together as a single string
    print("".join(result))

if __name__ == '__main__':
    print_common_characters()

```

	Input	Expected	Got	
✓	abcbde cdefghbb 3	bcd	bcd	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.